Machine Learning Practical Task

Naive Bayes - Sentiment Classification

Dataset: Sentiment Labelled Sentences
Source: UCI Machine Learning Repository **|** **Dataset Link:** Open Dataset
Data: Combined Amazon, IMDb, and Yelp labelled sentences
Target: 0 = Negative, 1 = Positive
Task: Build a Naive Bayes classifier to predict whether a sentence expresses positive or negative sentiment.

Field	Description
sentence	A short customer or movie review sentence
label	Sentiment class: 0 = Negative, 1 = Positive

Required Steps

1. Dataset Preparation
- Load and combine the Amazon, IMDb, and Yelp labelled sentence files.
- Display the dataset size and class distribution.

2. Text Preprocessing
- Check for missing values and duplicate sentences.
- Convert text to lowercase.
- Remove punctuation and special characters.
- Remove extra spaces.

3. Train/Test Split
- Use the sentence text as X and the sentiment label as y.
- Split the data into 80% training and 20% testing using stratification.

4. Text Representation
- Convert the sentences into numerical word-count features using CountVectorizer.

5. Naive Bayes Model
- Train a Multinomial Naive Bayes classifier.
- Use Laplace smoothing with alpha = 1.

6. Model Evaluation
- Evaluate the test set using Accuracy, Confusion Matrix, Precision, Recall, and F1-score.

7. New Sentences
- Classify these two new sentences:
- "The product is excellent and works perfectly."
- "The service was terrible and I will never come back."

Final Result

Report the final test accuracy and briefly compare the model performance on the Positive and Negative classes using Precision, Recall, and F1-score.